Supporting Information

Definitional Instability in Kinase Inhibitor Selectivity: An Empirical Characterization and Motivated Properties for a Well-Formed Measure

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This document reports, in full, four robustness checks summarized in the main text. All numbers are reproducible from the code repository (`https://github.com/polinavino/kinase-selectivity-definitions`); the script generating each table is named in its caption.

Table S1: Panel-size recoverability under alternative correlation bars

The main text calls a panel size "recovering" the full-panel ranking once mean Spearman ρ against the full panel exceeds 0.90, a convention borrowed from reliability research's "excellent agreement" bar rather than derived from the data (Methods, Results). Table S1 reports p^* (the smallest panel size reaching the bar) at four alternative bars, for both sparse-coverage datasets.

Table S1: Panel-size p^* (kinases) at four correlation bars. (`pstar_bar_sensitivity.py`)

Klaeger (343 kinases)	entropy	gini	s_score	ratio
bar = 0.80	80	260	50	320
bar = 0.85	80	260	80	320
bar = 0.90	110	290	110	343
bar = 0.95	140	320	200	343

Metz (172 kinases)	entropy	gini	s_score	ratio
bar = 0.80	50	50	50	170
bar = 0.85	50	50	50	170
bar = 0.90	80	80	50	170
bar = 0.95	110	110	80	170

The ratio remains the slowest to recover and the Gini coefficient the second-slowest for every bar tested; only the relative order of entropy and the S-score is bar-dependent, and only within $[0.80, 0.95]$.

Table S2: Bosc et al. window/ranking score and the CATDS proxy

Bosc et al.'s window score (WS) and ranking score (RS) and a CATDS-style proxy (Discussion) were computed on all four datasets to check whether they behave as additional members of the ratio family (WS, RS) or the distribution-based family (CATDS), as claimed in the Discussion. (`bosc_catds_comparison.py`)

Table S2: Spearman correlation of WS/RS against the ratio family, and of the CATDS proxy against the ratio, entropy, and Gini families.

Dataset	n_{cpd}	WS vs. ratio	RS vs. ratio	CATDS vs. ratio	CATDS vs. entropy / Gini
Davis	68	0.804	0.878	0.509	0.955 / 0.960
Klaeger	222	0.909	0.892	0.681	0.850 / 0.985
Anastassiadis	178	0.959	0.891	0.696	0.948 / 0.965
Metz	704	0.910	0.899	0.447	0.883 / 0.906

WS and RS correlate strongly with the ratio family on every dataset ($r = 0.80$–0.96), supporting their classification alongside $R(k)$. The CATDS proxy instead correlates more strongly with entropy and Gini than with the ratio on every dataset, supporting the Discussion's claim that CATDS is a distributional construction despite its historical classification as a ratio measure.

Table S3: Cross-definition correlations under the original metric operationalizations

Methods reports summary statistics for this check; the full 4×4 correlation matrices are given here. (`original_operationalization_robustness.py`)

Table S3: Davis ($n = 68$): manuscript S-score/Gini/ratio vs. literal Ka-weighted entropy (complete matrix).

	S-score	entropy(literal)	Gini	ratio
S-score	1.000	0.417	0.952	0.456
entropy(literal)	0.417	1.000	0.424	0.336
Gini	0.952	0.424	1.000	0.343
ratio	0.456	0.336	0.343	1.000

Klaeger ($n = 222$): manuscript S-score/Gini/ratio vs. literal Ka-weighted entropy (measured high-confidence entries only, per compound).

	S-score	entropy(literal)	Gini	ratio
S-score	1.000	0.570	0.722	0.223
entropy(literal)	0.570	1.000	0.688	0.789
Gini	0.722	0.688	1.000	0.616
ratio	0.223	0.789	0.616	1.000

Metz ($n = 704$): manuscript S-score/Gini/ratio vs. literal Ka-weighted entropy (measured entries only, per compound).

	S-score	entropy(literal)	Gini	ratio
S-score	1.000	0.239	0.879	0.211
entropy(literal)	0.239	1.000	0.416	0.758
Gini	0.879	0.416	1.000	0.229
ratio	0.211	0.758	0.229	1.000

Anastassiadis ($n = 178$): manuscript S-score/entropy/ratio vs. native %inhibition Gini (no baseline subtraction).

	S-score	entropy	Gini(native)	ratio
S-score	1.000	0.880	0.345	0.489
entropy	0.880	1.000	0.632	0.529
Gini(native)	0.345	0.632	1.000	0.314
ratio	0.489	0.529	0.314	1.000

On Klaeger and Metz, literal entropy correlates more strongly with the ratio than with either S-score or Gini – under this operationalization, entropy would not be classified in the distribution-based family at all on either sparse-coverage dataset. On Davis, literal entropy decouples from both families (consistent with Figure 1 of the main text). The native-Gini check on Anastassiadis

is the one case that preserves the qualitative two-family pattern, albeit more weakly than the manuscript's baseline-swept form.

Table S4: Candidate measure cross-dataset validation

Orientation (correlation with the number of active kinases, more negative is more correctly oriented toward "fewer active targets = more selective"), D3 robustness under the emphasis parameter β with the gate x_0 fixed (worst-case pairwise Spearman ρ), D3 robustness when x_0 itself is swept as a free parameter ("gate swept", the equivalent comparison to the entropy's baseline sensitivity; see Figure 6 of the main text), and the percentage of compounds for which D4 (monotonicity under a weak additional interaction) holds exactly. (`candidate_validation.py`)

Table S4: Candidate measure: orientation, D3 (two variants), and D4, across all four datasets.

Dataset	n_{cpd}	orientation r	D3, gate fixed	D3, gate swept	D4 (% compounds)
Davis	68	-0.969	0.998	-0.838	100.0
Klaeger	222	-0.936	0.999	-0.932	100.0
Anastassiadis	178	-0.936	0.997	$+0.913$	100.0
Metz	704	-0.791	0.991	-0.330	100.0

The "gate swept" column confirms, on three of four datasets, that the candidate's D3 satisfaction rests on fixing the gate x_0 by declaration rather than on the measure being inherently insensitive to its placement (worst-case ρ is far from 1, and on Klaeger and Davis is strongly negative). Anastassiadis is an exception ($\rho = +0.913$): on that dataset the candidate's ranking happens to be comparatively insensitive to the gate's placement as well, for reasons we do not have a mechanistic account of.

Smoothing-width T was also swept on Klaeger ($T \in \{0.5, 0.75, 1.0, 1.5, 2.0\}$): orientation and D3-emphasis-robustness are essentially unchanged across this range ($r \approx -0.94$, $\rho \geq 0.989$), and the worst-case pairwise ranking agreement across all five values of T is $\rho = 0.999$. Removing the gate inverts the measure's orientation entirely (gated $r = -0.936$ vs. ungated $r = +0.943$ against n_{active} on Klaeger), which is why the gate, not merely the smooth emphasis, is required for D1 and correct orientation jointly.